Frankie: Technical Status Report

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Executive Summary

Frankie Implementation - all of the features defined in the roadmap have been implemented and are used for organising the events. The roadmap contains a vague description of the features, increasing the risks of misunderstandings.

Mindsmiths Forge - used for implementing Frankie. It focuses on the ease of scaling the the number of users, while sacrificing the speed and ease of development. The platform is not sufficiently documented to enable the developers to use it independently from the Mindsmiths' developers.

Risks - only the developers from Mindsmiths know how to use the Forge platform. As long as Frankie is using Forge, a close collaboration with Mindsmiths is necessary. Replacing Forge means completely reimplementing Frankie.

Overview

The Frankfurt School request Serapion to conduct a technical assessment of the implementation of Frankie. The assessment of Frankie was conducted during a five day workshop in Zagreb, from March 6th to 10th 2023. This technical report contains the findings and the opinions on the status of the software implementation of Frankie, that were formed during this timespan.

The implementation of Frankie was guided by the roadmaps that were defined collaboratively between Mindsmiths and Frankfurt School (Ružica Pekić) during the course of the project. Based on the short description of the features that were defined in the roadmap, the evaluation of the expected deliverables was conducted. The existing implementation of Frankie delivers all of the aforementioned features which are used for inviting students based on their interests in the topics covered by the events.

The rapid prototyping approach to the development of Frankie led to the tradeoffs that resulted in low code quality, making further development ever more difficult. The development of further features could be sped up by refactoring and upgrading to the newest version of Mindsmiths Forge.

The technical documentation of Frankie is in good condition and addresses the most important aspects. Nevertheless, the documentation needs to be updated in several places to reflect recent changes in terms of modelling as well as infrastructure deployment.

Implementing Frankie using Mindsmiths Forge, a platform created and maintained by a startup, includes several potential risks. The complexity introduced by the platform is currently not beneficial for the efficiency of Frankie development. Additionally, the current state of development and documentation of the Mindsmiths Forge platform makes Mindsmiths the only company that has the necessary knowledge to efficiently use and operate it.

Adding new developers that are not related to the Mindsmiths will lead to considerable additional costs required for onboarding them onto the platform. Furthermore, the further development of Frankie using Forge will always require a close collaboration with Mindsmiths.

Frankie Implementation

Frankie is developed using Mindsmiths' platform called Mindsmiths Forge which is licensed based on the Platform License Agreement between Frankfurt School of Finance and Management GmbH, Ružica Pekić, and Mindsmiths d.o.o. signed on March 7th 2022.
Mindsmiths Forge is a development platform for creating, evolving, and running autonomous agents that support individuals based on their role within the project scope. This multi-agent platform provides a cloud solution making it highly available and scalable. The core components of Mindsmiths Forge are Mindsmiths' custom developed Rule Engine, three data stores (Mongo, Postgres, Redis), the custom developed component called Armory, Pulsar, a custom developed Dashboard and the custom developed component called Mitems. While the Rule Engine is evaluating the rules that represent the business logic, business flow as well as the screens that are displayed, the data stores persist temporary data (Signals) as well as persistent data (Facts). The Armory contains user facing components deciding what needs to be displayed and Mitems maps the actual content created on the Dashboard on the components and templates in Armory. Finally, Pulsar is a messaging and streaming framework that connects all the described components. This infrastructure is deployed on Microsoft Azure and uses Kubernetes for infrastructure management.

The adaptation of the platform to implement the use case of Frankie is created by Mindsmiths employees based on a the Contract Of Work between Frankfurt School of Finance and Management GmbH, Ružica Pekić, and Mindsmiths d.o.o. signed on March 7th 2022.

The development followed an agile approach based on sprints. In that setting, Ružica Pekić acted as the customer, whereas Mindsmiths d.o.o. provided the product owner as well as the development team. The initial backlog was defined in the aforementioned Contract Of Work and was later replaced by a roadmap, which was adjusted based on mutual consent among the contract partners. The goals of the sprints were defined based on the presented opportunities at the start of each sprint. The rapid prototyping approach, supported by all parties, led to a focus on functionality with a reduced focus on code quality and documentation. To track the progress in the project, Frankie Progress Report documents were created to summarize the progress made after each sprint. Nevertheless, these documents do not cover the reasoning and decision making that was done in the project, which would be beneficial for future activities.

Roadmap vs. Current Status

The development tasks handed over to the development team were underspecified. This can be attributed to the fact that the roadmap was used as a development planning tool while missing further specifications. In this context, the only possible comparison between what was planned and the as-is status is the comparison between the roadmap and the artefacts that were delivered. The following table summarizes the artefacts created during each phase of the roadmap. The documents referred to in the table can be found in Frankie's google drive[1].

Roadmap	Requested	Delivered Artifacts
Definition	• Define target population	• Target populations definitions (e.g. 3_FSFM phase 1 report_december 2021.pdf)
	• Define event types	• Event types definitions and modeling (see Frankie Technical Documentation)
	• Define data sources	• Evaluation of datasources: Canvas, Full Fabric and CLM

[1] This Google Drive is owned by Mindsmiths. For access to this or any other referenced resources, please contact Ružica Pekić.

	• Define communication channels	• Setup of communication channels
Prototype	• Platform setup	• Mindsmiths Forge setup for Frankie (see Frankie Infrastructure Documentation)
	• FS email integration	• Feature: Email integration with Frankfurt School
	• Event model	• Definition and implementation of event model (see Frankie Technical Documentation: 2.2.1 Event content management) ○ Feature: Event model
	• Organizer model	• Definition and implementation of organizer model (see Frankie Technical Documentation: 2.2.4 Organizer) ○ Feature: Organizer model
	• Attendee model	• Definition and Implementation of attendee model (see Frankie Technical Documentation: 2.2.5 & 2.2.6 Attendee types & statistics) ○ Feature: Attendee model
	• Data import	• Data import of initial users
	• Basic matchup	• Feature: Matchup mechanism using event tags and invitee's interests • See Frankie Technical Documentation: 2.4.3 Basic matching algorithm
	• Invitation flow	• Definition and implementation of invitation flow (see Frankie Technical Documentation: 2.2.7 Event invite, 2.7 Invitation flow breakdown) ○ Feature: Invitation flow for attendees
	• Feedback flow	• Definition and implementation of feedback flow (see Frankie Technical Documentation: 2.2.8 Event feedback) ○ Feature: Feedback flow for attendees
	• Admin interface	• Feature: Dashboard ○ Feature: Admin interface (see Dashboard) ○ see Frankie Technical Documentation: 2.2.10 Admin dashboard
	• Testing and bug fixing	• Not trackable
	• Prototype release	• Released prototype
	• Branding	• Creation of design suggestions (created by Sawraft)

Beta v1.0	• Design elements	• Creation of design elements based on selected design (created by Sawraft)
	• UI elements	• Translation and implementation of design into UI elements
	• Canvas integration	• Feature: Canvas Integration ○ Code for Canvas integration is available, but was later dropped since Canvas was not ready to be used
	• Attendee onboarding	• Attendee onboarding (see Frankie Technical Documentation: 2.3 Onboarding) • Feature Onboarding flow • Feature: Shadow onboarding
	• Organizer onboarding	• Organizer onboarding (see Frankie - Guidelines for Event Organizers)
	• Event creation flow	• Definition of event creation flow (see Frankie Technical Documentation: 2.6 Event creation, Frankie User Journey) • Feature: Event creation via dashboard
	• Feedback flow	• Definition of feedback flow (see Frankie Technical Documentation: 2.2.8 Event feedback, Frankie User Journey) • Feature: Email and feedback collection after events
	• Testing and bug fixing	• Not trackable
	• Beta v1.0 release	• Released Beta v1.0
Beta v1.1	• Additional reminders	• Feature: Sending additional event reminders
	• Ad hoc messages	• Feature: Sending ad-hoc messages to invitees
	• Attendee model v2	• Updated attendee model (see Frankie Technical Documentation)
	• Analytics 1.1	• Definition of analytics (Number of value creating moments, Number of Events, Attendance per Event, Attendee satisfaction) ○ Feature: Tracking email openings ○ Feature: Monitoring events for organizers
	• Cloud sheet as a new CRM	• Google sheet as a pseudo CRM (see Frankie CRM) • see Frankie Technical Documentation: 2.5 Google Sheets integration

	• Data migration	• Feature: Integration of the google sheet as the source of truth
	• Testing and bug fixing	• Released Beta v1.1
	• Beta v1.1 release	• Not trackable
Beta v1.2	• User interests definition	• Definition of selectable interests
	• Onboarding flow	• Updated onboarding flow based on new landing page
	• New email and domain	• Integration of new Frankie email and domain
	• Cloud sheet CRM 1.2	• New Frankie CRM google sheet (without students courses and semesters, new tags)
	• Signup flow	• Onboarding / Signup flow (see Frankie Technical Documentation: 2.3 Onboarding & 2.7 Invitation flow breakdown) ○ Feature: Information of signed up people can be updated using the onboarding flow
	• Who is Frankie content v1	• Who is Frankie description (see Who is Frankie)
	• Simple landing page v1	• Feature: Landing page
	• Referral invitation flow	• Definition of referral invitation flow (Dropped due to landing page)
Beta v1.3	• Tickets system architecture	• Definition of ticket architecture and flow (see Frankie Technical Documentation: 2.13 Ticketing system)
	• Tickets prototype	• Feature: Ticketing system • Feature: QR code scanner
	• Attendance analytics	• Feature: Analytics of attendance
	• Testing and bug fixing	• Not trackable

Risks

Although the requested features have been implemented and functional, several risks were identified in the current project setting; some of which are connected to the current project structure and some related to the technical development.

The first set of risks is related to the planning, execution and reviews of sprints. While Frankie Progress Reports represent, to some degree, a review process, an actual review meeting for achieved results is missing. This may lead to a misunderstanding of requested vs created artefacts within the Frankie team.
In a similar context, the planning of sprints was based on the presented opportunities and needs, which would find their way into the roadmap. This roadmap only names tasks that need to be done until the next release, without a description, clarification of their content, acceptance criteria, priorities and without having sprint scopes in mind. Therefore, it is impossible to evaluate if the created features match the expectations of the project lead.
Third, due to a limited visibility of the necessary development tasks, it is hard for anyone except the developers themselves to track progress. This prevents the project lead from adjusting the capacity of the team based on the progress speed. Furthermore, the lack of the visibility of the development tasks leads to misunderstandings about the speed of development progress.

The second set of risks is related to the technical implementation and in particular the Mindsmiths Forge platform.
First, the platform is very flexible, powerful and can implement a huge variety of use cases. Consequently, the platform introduces a layer of complexity that gives it the aforementioned capabilities. Given the current features of Frankie, this complexity is not beneficial. This results in two issues: On the one hand, new features, that the platform was not inherently designed for, take more time to implement.
Second, due to a lack of documentation of the platform, its complexity makes it very difficult for developers from outside of Mindsmiths to be onboarded to the platform and take over development tasks. Furthermore, in case the developers from Mindsmiths become unavailable and new developers have to take over the project, new components can only be deployed in close collaboration with Mindsmiths. This can result in a development bottleneck and lock the platform vendor within Frankie.
Third, the current status of Frankie's development is tightly coupled to the Mindsmiths Forge. If the decision is made to transition the current prototype to a product without using the platform, the software solution needs to be recreated from scratch.